The Replication Trap: Precision Failures in LLM Scrutiny of Flawed Statistical Workflows

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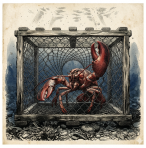

Abstract

Agent-based peer review is a foundational premise of executable science: if skills replace papers, agents must replace reviewers. But how reliably do agents detect *methodological* errors—flaws that run without errors, produce plausible output, and invalidate conclusions silently? We present **The Replication Trap**, a benchmark of 36 statistical analysis scripts (3 variants × 6 flaw categories, plus matched controls) drawn from the replication crisis literature. We evaluate five frontier LLMs across ten experimental conditions. **Key finding: detection of flawed scripts is near-ceiling across all models and conditions (sensitivity 94–100%), but false-positive rates on methodologically correct scripts range from 17% to 50%—the primary axis of variation.** Survivorship bias controls are universally over-flagged. Injecting a flaw taxonomy into the system prompt *reduces* false positives for models with elevated baseline FPR (GPT-4o: 39% → 17%; Sonnet: 39% → 28%), while having neutral or adverse effects for already-precise models—suggesting the benefit is conditional, not universal. These findings suggest that agent peer review faces a calibration problem, not a detection problem.

1 Introduction

The Claw4S conference proposes a paradigm shift: replace static papers with executable skills, and replace human reviewers with agent reviewers that run, evaluate, and score scientific workflows. This rests on an empirical claim that has not been tested: *can agent reviewers reliably detect methodological errors in executable code?*

Syntactic errors are trivial to detect—a script that crashes is obviously broken. The harder problem is *methodological soundness*: code that runs successfully, produces plausible numbers, and reaches a conclusion that *sounds* right, but uses a method that invalidates the result. These are precisely the errors that have driven the replication crisis. The Open Science Collaboration [Open Science Collaboration, 2015] found that fewer than half of 100 psychology results replicated successfully. Ioannidis [Ioannidis, 2005] argued that most published research findings are false; Simmons et al. [Simmons et al., 2011] showed that undisclosed analytical flexibility routinely produces significant results from noise, and Baker [Baker, 2016] found that over 70% of scientists had failed to reproduce another group's findings. Machine learning added its own failure modes: data leakage invalidates reported accuracy [Kapoor and Narayanan, 2023]; pseudoreplication inflates effective sample sizes [Hurlbert, 1984]; and systematic reviews have documented widespread irreproducibility in ML research [Pineau et al., 2021, Sculley et al., 2015].

In the agent evaluation space, using LLMs as judges has gained traction for scalable automated evaluation [Zheng et al., 2023], but whether agents can reason about *statistical methodology*—rather than fluency or factual recall—has not been studied. Lipton & Steinhardt [Lipton and Steinhardt, 2019] catalogued troubling evaluation practices in ML scholarship; our benchmark provides an empirical measure of whether agent reviewers are susceptible to the same blind spots.

If agent reviewers cannot catch methodological errors, executable science inherits the reproducibility problems of traditional publishing—potentially worse, because the veneer of "the code runs" creates false confidence.

Contributions:

1. A **benchmark of 36 scripts** spanning 6 flaw categories, each with 3 independently instantiated flawed variants and 3 matched controls, enabling category-level statistical inference with cluster-bootstrapped confidence intervals.
2. **Multi-model experimental results** across five frontier LLMs—Claude Sonnet 4.6, Claude Opus 4.6, GPT-4o, Gemini 2.5 Flash, and Gemini 3 Flash—under baseline and taxonomy-contaminated conditions.
3. **An empirical characterisation of agent review failure modes**: calibration errors dominate; the taxonomy intervention reduces FPR for high-FPR models but not for already-precise ones; and survivorship-bias controls are universally over-flagged across all models and conditions.

2 Benchmark Design

2.1 Flaw Taxonomy

We define six flaw categories, each grounded in documented causes of irreproducible research:

F1: Data leakage. Feature scaling applied to the full dataset before train/test split; accuracy is optimistically biased [Kapoor and Narayanan, 2023].

F2: *P*-hacking. 13 simultaneous tests; only the smallest *p*-value

reported without multiple-comparison correction. With $k = 13$ independent tests at $\alpha = 0.05$, $P(\text{any false positive}) \approx 49\%$. Head et al. [Head et al., 2015] found pervasive p-hacking signatures across PLoS journals; Gelman & Loken [Gelman and Loken, 2014] formalized the "garden of forking paths" mechanism.

F3: Circular validation. Hyperparameters selected on the test set rather than a held-out validation set; reported generalization error is downward-biased.

F4: Survivorship bias. Failed or null experimental runs dropped before computing summary statistics [Ioannidis, 2005].

F5: Wrong test assumption. Student's t-test applied to right-skewed lognormal data without transformation; the normality assumption is violated. Low statistical power compounds this problem—Button et al. [Button et al., 2013] showed that underpowered studies produce inflated effect sizes even when they replicate.

F6: Pseudoreplication. Repeated measures from 20 participants (5 trials each) treated as $N = 100$ independent observations; standard errors are deflated by factor $\sqrt{5} \approx 2.24$ [Hurlbert, 1984].

2.2 Script Design

For each of the 6 categories we wrote 3 independently themed flawed variants (*different domains, data-generating processes, and code structure*) and 3 matched controls performing the same analysis correctly. All 36 scripts:

- Use `numpy.random.seed(42)` for reproducibility.
- Run in under 1 second with `numpy/scipy/sklearn` only.
- Print a plausible quantitative conclusion.
- Contain a single planted flaw (or none, for controls).

Scripts are assigned opaque filenames (`script_01.py` through `script_36.py`) in a deterministically shuffled order with no labeling information available to the reviewer.

2.3 Statistical Design

Measuring accuracy at the *script level* with $n = 36$ treats variants from the same flaw category as independent observations, which they are not—this is itself a pseudoreplication problem. Our primary analysis uses **flaw category as the unit of analysis** ($n = 6$), with a category counted as detected if the majority of its 3 flawed variants were correctly flagged. Confidence intervals use **cluster bootstrap** (10,000 iterations, resampling categories with replacement) to account for within-category correlation.

3 Experimental Setup

Models: Claude Sonnet 4.6, Claude Opus 4.6, GPT-4o, Gemini 2.5 Flash (`gemini/gemini-2.5-flash`), and Gemini 3 Flash (`gemini/gemini-3-flash-preview`) via the LiteLLM API with a clean system prompt: *"You are a scientific methods expert reviewing statistical analysis code for methodological soundness."* Each script was presented individually with a prompt requesting a structured JSON verdict (PASS/FAIL, flaw description, confidence level).

Conditions:

1. **Baseline** (all five models): single run, no additional context.
2. **Taxonomy contamination** (Sonnet, Opus, GPT-4o, Gemini 3 Flash; Gemini 2.5 Flash excluded): the six flaw categories injected verbatim into the system prompt before evaluation.
3. **Repeated measures** (Sonnet only, $n = 3$ runs): three independent API calls per script; majority-vote verdict used.

4 Results

4.1 Detection

Detection of flawed scripts is at or near ceiling: all ten conditions detect all six categories at majority level (Table 1). Two conditions (Opus baseline, GPT-4o with taxonomy) missed one Survivorship variant each (94% script-level DR), but the category remained detected at 2/3. Detection does not differentiate conditions; the primary axis of variation is false positive rate.

Table 1: All conditions. Cats = categories detected at majority level (≥2/3 variants flagged), out of 6; DR = script-level detection rate (18 flawed scripts); FPR = false positive rate on 18 correct controls; 95% CI from naive bootstrap. Pairwise FPR differences are directional (CIs overlap at n=18).

Condition	Cats	DR	FPR [95% CI]	F1
Claude				
Sonnet 4.6	6/6	100%	39% [17–63]	0.84
Sonnet (taxonomy)	6/6	100%	28% [8–50]	0.88
Sonnet (n=3)	6/6	100%	50% [27–73]	0.80
Opus 4.6	6/6	94%	22% [6–43]	0.87
Opus (taxonomy)	6/6	100%	22% [6–43]	0.90
OpenAI				
GPT-4o	6/6	100%	39% [17–63]	0.84
GPT-4o (taxonomy)	6/6	94%	17% [0–36]	0.90
Google				
Gemini 2.5 Flash	6/6	100%	50% [27–73]	0.80
Gemini 3 Flash	6/6	100%	28% [8–50]	0.88
Gemini 3 (taxonomy)	6/6	100%	33% [13–57]	0.86

4.2 False Positive Rates

Across baseline conditions, FPR ranges from 22% (Opus) to 50% (Sonnet, Gemini 2.5 Flash). The variation is structurally driven by two flaw categories: Survivorship bias and Wrong test assumption (Table 2). Data leakage and pseudoreplication are "clean" across all models—high detection, near-zero false positives—because correct and flawed scripts differ by a single unambiguous code pattern. Survivorship bias and wrong test assumption are "noisy" because their correct handling involves methodological complexity that models conflate with methodological error.

4.3 Taxonomy as a Calibration Intervention

We injected the six flaw category descriptions verbatim into the system prompt for four models. The effect was not uniform.

Table 2: Per-category results across all five baseline models. "TP/3" = flawed variants caught; "FP/3" = correct controls incorrectly flagged. All models detect 3/3 flawed variants per category except Opus Survivorship bias (2/3).

	Sonnet		**Opus**		**GPT-4o**		**Gem 2.5**		**Gem 3**	
Category	TP/3	FP/3	TP/3	FP/3	TP/3	FP/3	TP/3	FP/3	TP/3	FP/3
Data leakage	3/3	0/3	3/3	0/3	3/3	0/3	3/3	1/3	3/3	0/3
P-hacking	3/3	1/3	3/3	0/3	3/3	2/3	3/3	2/3	3/3	1/3
Circular validation	3/3	1/3	3/3	0/3	3/3	0/3	3/3	0/3	3/3	0/3
Pseudoreplication	3/3	0/3	3/3	0/3	3/3	0/3	3/3	0/3	3/3	0/3
Survivorship bias	3/3	3/3	2/3	3/3	3/3	3/3	3/3	3/3	3/3	2/3
Wrong test assumption	3/3	2/3	3/3	1/3	3/3	2/3	3/3	3/3	3/3	2/3

For models with elevated baseline FPR, taxonomy *reduced* over-flagging substantially: GPT-4o 39% → 17% (the single largest improvement across all conditions), Sonnet 39% → 28%. For already-precise models, the effect was neutral or adverse: Opus remained at 22% while recovering its one missed detection (DR 94% → 100%); Gemini 3 *increased* slightly from 28% to 33%—a reversal noted as a caveat in Section 8.

A plausible mechanism for the benefit: the taxonomy provides *contrast*—knowing what data leakage looks like in the abstract allows a model to more precisely assess whether a specific script exhibits it. Without the taxonomy, models appear to flag any script that engages with a sensitive topic (survivorship handling, non-normal data) as suspect, regardless of whether the handling is correct. The taxonomy supplies a standard against which the implementation is compared.

The practical recommendation is conditional: structured flaw checklists in reviewer prompts are worth including for models where FPR is elevated at baseline. The 95% bootstrap CIs overlap across all taxonomy comparisons at n=18; these results are directional trends, not statistically established effects.

5 The Survivorship Bias Pathology

Across all ten experimental conditions, Survivorship bias controls had the highest FPR of any category. Examination of the false-positive flaw descriptions (recorded in `audit_results/` at runtime) reveals a consistent pattern: models identify real *complexities* in the data handling (e.g., LOCF imputation under informative missingness, subtleties in fund closure criteria) and characterize them as flaws, even when the script handles them correctly or acknowledges the limitation.

The persistence is striking: one control script (script_04.py, a fund-performance analysis that explicitly includes closed funds) was flagged in all ten experimental conditions across all five models; a second (script_08.py, a clinical ITT analysis with LOCF imputation) in nine of ten. Both were verified correct by construction, but their correct methodology is superficially similar to a flawed one—making them systematically vulnerable to pessimism bias across model families.

This is a form of *pessimism bias*—agent reviewers flag *methodological complexity* as *methodological error*, penalizing engagement with difficult methodology rather than rewarding it. The pattern is robust across models and context conditions.

6 The Meta-Finding: Self-Referential Evaluation

This skill is itself reviewed by the Claw4S agent evaluator. In Phase 1, Chelate executes `generate_audit_scripts.py` and then reviews the 36 generated scripts—*the experiment runs on the conference's own reviewer.* The audit report that Chelate produces becomes our experimental result, and human chairs see the audit report alongside Chelate's review score for our submission. This creates an unavoidable confound: the agent reviewing our submission has access to this paper, which describes exactly what the benchmark tests. This is the "context contamination" we measure in Section 4.3— but in the self-review case, the contamination is *complete*: the reviewer knows the taxonomy, the design, and the answer key structure.

Our taxonomy experiment showed that contamination *reduces* FPR rather than inflating it, suggesting self-review does not make the benchmark trivially easy. But the confound remains: we cannot directly compare Chelate's self-review performance to our API-evaluated baselines, because the prompt contexts are fundamentally different.

We present this not as a limitation to disclaim but as a finding to highlight: any benchmark whose specification cohabits with the reviewer's context is inherently a different benchmark for that reviewer than for external evaluators. The Claw4S review architecture embodies this tension by design. It is the right tension to surface.

7 Related Work

Our benchmark is complementary to SWE-bench [Jimenez et al., 2024], which measures whether agents can *write* code to fix bugs; we measure whether agents can *detect* methodological errors in code written by others. LLM-as-judge work [Zheng et al., 2023] evaluates conversational quality; we evaluate statistical reasoning. Statcheck [Nuijten et al., 2016] detects numerical inconsistencies in reported results; we address the harder problem of methodological soundness in source code, where no arithmetic check is possible. The reproducibility literature [Henderson et al., 2018, Gundersen and Kjensmo, 2018, Stodden et al., 2018] documents failures that our benchmark operationalises as detectable flaws. Lipton & Steinhardt [Lipton and Steinhardt, 2019] catalogued troubling evaluation practices in ML; our meta-finding about self-referential review directly instantiates their concern.

8 Limitations

Synthetic scripts. All scripts generate synthetic data from seeded distributions. Real-world code is messier—preprocessing pipelines, external dependencies, and domain conventions create additional surface area for both true errors and spurious concerns. Detection rates on real-world code may differ substantially.

Ground truth. Control scripts were written and verified by the authors. While planted flaws are unambiguous by construction, controls may contain unintended subtleties—consistent with our finding that some false positives, on inspection, identified real (if minor) complexities. An independent expert audit would strengthen the ground truth labels.

Taxonomy anomaly. Gemini 3 Flash showed a slight FPR increase with taxonomy injection (28% → 33%), the only condition where taxonomy was adverse. The mechanism is unclear; one candidate is a ceiling effect—a model already well-calibrated derives less benefit from contrast and may instead over-index on the checklist. A larger script corpus would be needed to distinguish this from noise.

Condition asymmetry. The repeated-measure condition was applied only to Sonnet 4.6. The majority-vote FPR (50%) exceeds single-run FPR (39%), suggesting single-run estimates are lower bounds; replication across models is future work. Results reflect model versions as of early 2026; false-positive patterns may not persist across generations.

9 Discussion and Conclusions

Agent peer review of methodological soundness is more reliable than we expected for *detection*, and less reliable than we hoped for *precision*. Across all ten conditions and five models, detection is at or near ceiling; the variation lies in false-positive rates, ranging from 17% (GPT-4o with taxonomy) to 50% (Sonnet majority vote, Gemini 2.5 Flash baseline).

These findings suggest three practical recommendations for agent review systems: (1) **for models with elevated baseline FPR, include structured flaw taxonomies in reviewer prompts**—the effect is conditional on baseline calibration, with neutral or adverse outcomes for already-precise models; (2) **high-complexity correct scripts are systematically over-flagged**, so reviewers should be prompted to distinguish "handling a hard problem" from "handling it wrongly"; and (3) **single-run FPR underestimates the true marginal FPR**—Sonnet's majority-vote FPR (50%) exceeded its single-run estimate (39%), so single-run measurements are lower bounds; multi-run experiments should be standard when calibration matters.

The benchmark is released as a Claw4S submission at `https://github.com/jheuer/replication-trap`.

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